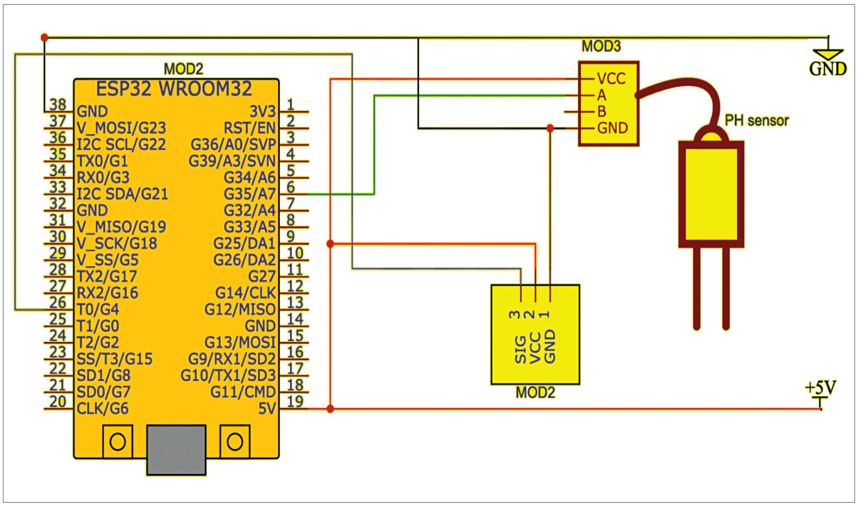

Fig. 8: Circuit diagram of the proposed system

on during deep sleep mode when using ext0 as a wakeup source.

Software

The code is prepared in Arduino IDE. Here, the PIN is defined for the sensor, and then the pH value is extracted using a formula. In the setup function, the sensor data is processed with the sleep function to enable the processor to sleep and wake on time to process the sensor data. Fig. 5 shows a snippet of the source code.

Fig. 6 represents additional information about the setup and the components used. The pinout of the sensing probe, along with the probe assembly, provides access to the pH readings, taking into account the adjustment of offsets due to analogue readings and pH limits. Table 2 displays the bill of materials, and Table 3 shows the pin connections for interfacing the sensor.

Circuit diagram

The circuit is built around the touch sensor (MOD1), ESP32 (MOD2), pH sensor (MOD3), and a few jumpers. Fig. 7 illustrates the wiring of the proposed setup on the breadboard, and Fig. 8 shows the circuit diagram of the proposed system.

Before assembling the project, refer to Fig. 7 and 8. After proper assembly of the circuit based on the circuit diagram, upload the code to ESP32 and then test the device by powering it. You will receive the pH value in the serial monitor when you dip the sensor into the liquid to be measured.

Note. In conclusion, it is possible to measure the pH of a hydroponic solution using a sensor while conserving power by operating it in deep sleep mode. EFY

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*


Dr Geetali Saha is a faculty member at the Department of Electronics and Communication Engineering, GCET, Anand, Gujarat, and Dhruv Vachhani, a student coordinator from the 3rd Year CSE - IoT Department, is working on this sponsored project with Dr Saha